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Lecture 1: Optimal Transport

Lecture 2: Generative Flows

Lecture 3: From 1D Processes to Flows

Lecture 4: Bayesian Inverse Problems

Berlin Mathematics Research Center MATH+

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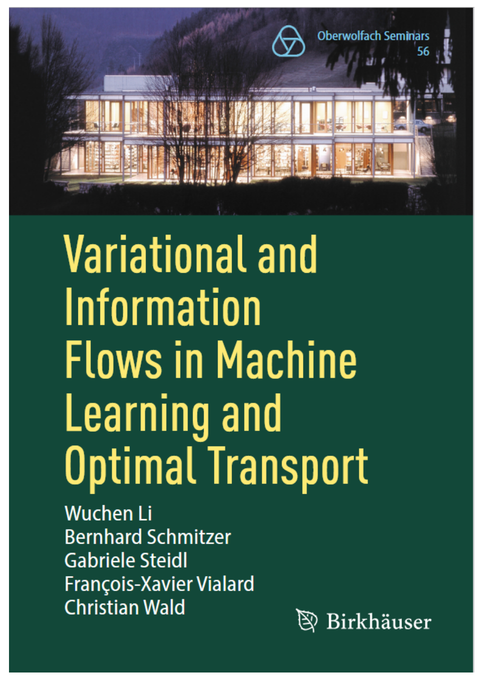
2. Generative Flows

1. Basic Idea of Generative Flows
2. Absolutely Continuous Curves in $(\mathcal{P}_2(\mathbb{R}^d), W_2)$
3. Flow Matching
4. A Glimpse to Score-based Diffusion



Refs: Wald/St.: Flow Matching: Markov kernels, transport plans and stochastic processes, MFO Seminar Series, Springer 2025

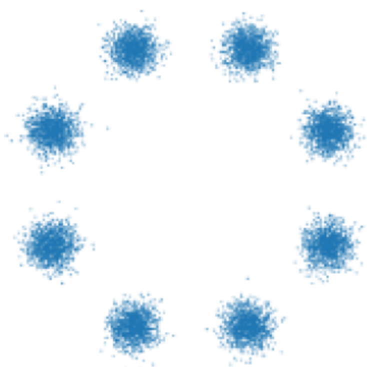
1. Basic Idea of Generative Flows

Aim: Sampling from a probability distribution μ_X (target distribution) given:

- some samples (oft many = data hungry)
- probability density function $p_X(x) = \exp(-\psi(x))/C$ (Boltzmann density) with or without normalizing factor C

It is only easy to sample from

- 1d distributions (cdf)
- $d \gg 1$ uniform or Gaussian "friendly" distributions μ_Z (latent distribution)



$d = 2$

$d = 28^2 = 784$

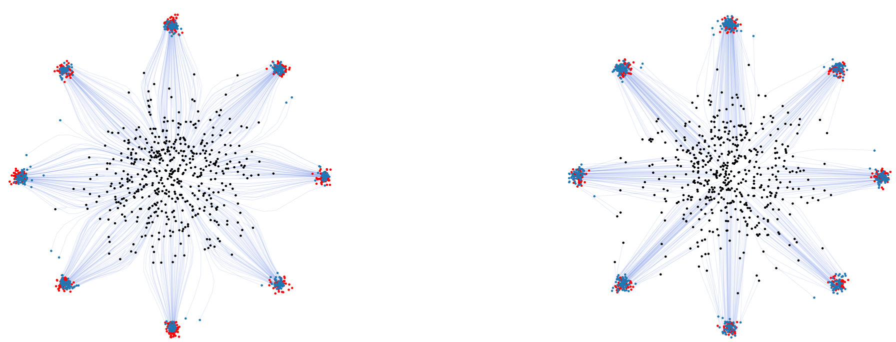
Refs: J. Chemseddine, C. Wald, R. Duong, St: Neural sampling from Boltzmann densities: Fisher-Rao curves in the Wasserstein geometry, ICLR 2025

Flow Based Generative Models

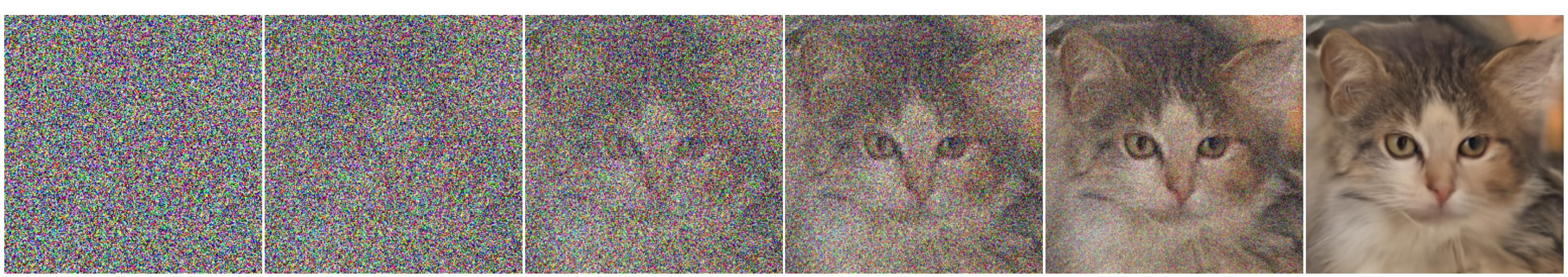
Idea: Learn a curve from μ_Z to μ_X

Basic methods:

- score-based diffusion models (Sohl-Dickstein 2015; Song/Ermon et al. 2019)
- flow matching (Lipman et al. 2022/23, Liu et al. 2022/23, Albergho et al. 2023)
- consistency models, e.g. inductive moment matching (Zhou et al. ICLM 2025, Boffi et al. 2025)



Trajectories of points via flow matching from different couplings, $d = 2$



Single trajectory for a flow matching model trained on cat images, $d = 256^2$

2. Generative Flows

In a complete metric space (X, d), a curve $\gamma : I \to X$ is called absolutely continuous ($AC^2(\mathbb{R}^d)$), if there exists as $g \in L_2(I)$ such that

$$\mathrm{d}(\gamma(s), \gamma(t)) \leq \int_s^t g(r)\,\mathrm{d}r \quad \text{for} \quad s \leq t,\ s, t \in I.$$

Then there exists a.e. the metric derivative (speed) which is indeed the smallest function g in the above inequality

$$|\gamma'|(t) := \lim_{s \to t} \frac{\mathrm{d}(\gamma(t), \gamma(s))}{t - s}.$$

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27

Absolutely Continuous Curves in $(\mathcal{P}_2(\mathbb{R}^d), W_2)$

- A narrowly continuous curve $\mu_t : I = [0,1] \to \mathcal{P}_2(\mathbb{R}^d)$ absolutely continuous, if there exists a Borel measurable vector field $v : I \times \mathbb{R}^d \to \mathbb{R}^d$ with $\|v_t\|_{L_2(\mu_t)} \in L_2(I)$ such that (μ_t, v_t) satisfies the continuity equation (CE)

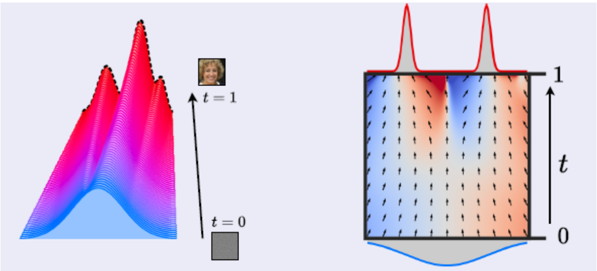
$$\partial_t \mu_t + \nabla_x \cdot (\mu_t v_t) = 0$$

- If μ_t has a smooth density p_t, then

$$\partial_t p_t + \nabla_x \cdot (p_t v_t) = 0$$

- otherwise (CE) is meant in a weak sense, i.e. for all $\varphi \in C_c^\infty((a,b) \times \mathbb{R}^d)$:

$$\int_I \int_{\mathbb{R}^d} \partial_t \varphi \, \mathrm{d}\mu(t) \mathrm{d}t + \int_I \int_{\mathbb{R}^d} \langle \nabla_x \varphi, v \rangle \, \mathrm{d}\mu_t(x) \mathrm{d}t = 0$$



Absolutely Continuous Curves in $(\mathcal{P}_2(\mathbb{R}^d), W_2)$

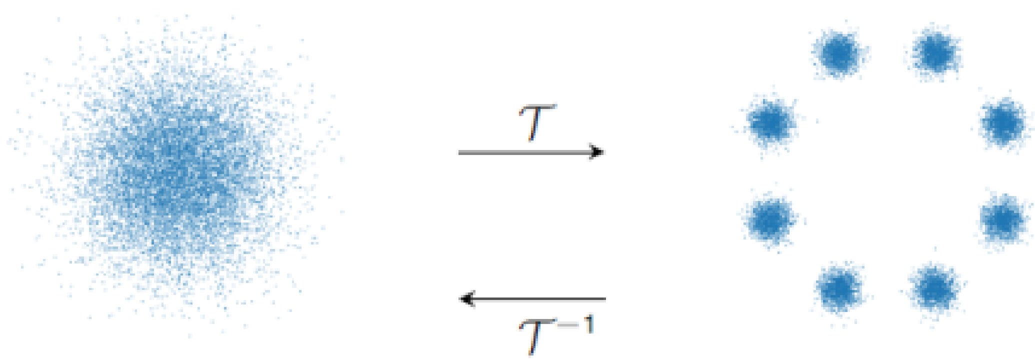
- Let (μ_t, v_t) fulfill (CE) and

$$\int_I \sup_{x \in B} \|v_t(x)\| + \mathrm{Lip}(v_t, B)\,\mathrm{d}t < \infty \quad \forall \text{compact } B \subset \mathcal{B}(\mathbb{R}^d).$$

Then there exists a solution $\phi : I \times \mathbb{R}^d \to \mathbb{R}^d$ of the ODE

$$\partial_t \phi(t,x) = v_t(\phi(t,x)), \quad \phi(0,x) = x$$

and $\phi(t,\cdot)_\sharp \mu_0 = \mu_0 \circ \phi(t,\cdot)^{-1} = \mu_t$.



Also the other direction is true under some conditions.

Assume that ϕ satisfies the ODE for some locally bounded vector field v_t on $I = (a, b)$. For $\varphi \in C_c^\infty((a, b) \times \mathbb{R}^d)$, we can compute

$$
\begin{aligned}
0 &= \varphi(b, \phi(b, x)) - \varphi(a, \phi(a, x)) \\
&= \int_{\mathbb{R}^d} \varphi(b, \phi(b, x)) - \varphi(a, \phi(a, x)) \mathrm{d}\mu_0 \\
&= \int_{\mathbb{R}^d} \int_a^b \frac{\mathrm{d}}{\mathrm{d}t}(\varphi(t, \phi(t, x)))\, \mathrm{d}t\mathrm{d}\mu_0 \\
&= \int_{\mathbb{R}^d} \int_a^b \langle \nabla_x \varphi\big(t, \phi(t, x)\big), v_t\big(t, \phi(t, x)\big)\rangle + (\partial_t \varphi)\big(t, \phi(t, x)\big)\, \mathrm{d}t\mathrm{d}\mu_0 \\
&= \int_a^b \int_{\mathbb{R}^d} \langle \nabla_x \varphi, v_t\rangle + \partial_t \varphi \,\mathrm{d}[\phi(t, \cdot)_\sharp \mu_0]\mathrm{d}t
\end{aligned}
$$

Thus $\mu_t := \phi(t, \cdot)_\sharp \mu_0$ satisfies (CE) (in a weak sense).

Summary

- A narrowly continuous curve $\mu_t : I = [0,1] \to \mathcal{P}_2(\mathbb{R}^d)$ absolutely continuous, if there exists a Borel measurable vector field $v : I \times \mathbb{R}^d \to \mathbb{R}^d$ with $\|v_t\|_{L_2(\mu_t)} \in L_2(I)$ such that (μ_t, v_t) satisfies the continuity equation (CE)

$$\partial_t \mu_t + \nabla_x \cdot (\mu_t v_t) = 0$$

- Let (μ_t, v_t) as above and

$$\int_I \sup_{x \in B} \|v_t(x)\| + \mathrm{Lip}(v_t, B)\, \mathrm{d}t < \infty \quad \forall \text{compact } B \subset \mathcal{B}(\mathbb{R}^d).$$

  Then there exists a solution $\phi : I \times \mathbb{R}^d \to \mathbb{R}^d$ of the ODE

$$\partial_t \phi(t,x) = v_t(\phi(t,x)), \quad \phi(0,x) = x$$

  and $\phi(t,\cdot)_\sharp \mu_0 = \mu_0 \circ \phi(t,\cdot)^{-1} = \mu_t$.

- If a nice person gifts you such a velocity field v_t, then use your favorite ODE solver to transport samples from $\mu_0 = \mu_X$ to samples of $\mu_1 = \mu_Z$ or backwards using $-v_{1-t}$.

All you need is v_t !

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How to Learn a Velocity Field?

Task: Learn/approximate v_t by neural network v_t^θ!

Loss function for learning:

$$\mathcal{L}(\theta) := \mathbb{E}_{t\sim\mathcal{U}(0,T),x\sim\mu_t}\Big[\big\|v_t^\theta(x) - v_t(x)\big\|^2\Big],$$

not accessible.

Idea: Make it conditional since up to a constant

$$\mathcal{L}(\theta) = \mathbb{E}_{t\sim\mathcal{U}(0,T),\, y\sim\mu_1,\, x\sim\mu_t(\cdot|y)}\Big[\big\|v_t^\theta(x) - v_t^y(x)\big\|^2\Big]$$

We will see that conditional velocity fields v_t^y belonging to a conditional flow $\mu_t(\cdot|y)$ starting in δ_y are available in an analytic form in certain cases e.g. from couplings of μ_X and μ_Z (optimal transport) as we consider next

All you need is conditional v_t^y !

Rewriting the Loss Function

Random variables: $X_0 \sim P_{X_0} = \mu_0$, $X_1 \sim P_{X_1} = \mu_1$, $X_t \sim P_{X_t} = \mu_t$, $\mu_t(\cdot|y) = P_{X_t|X_1=y}$

$$P_{X_t,X_1} = P_{X_t|X_1=y} \times_y P_{X_1} = P_{X_1|X_t=x} \times_x P_{X_t}$$

with disintegration of P_{X_0,X_1}, i.e.

$$\int_{\mathbb{R}^d \times \mathbb{R}^d} f(x,y)\,\mathrm{d}P_{X_t,X_1}(x,y) = \int_{\mathbb{R}^d} \int_{\mathbb{R}^d} f(x,y)\,\mathrm{d}P_{X_1|X_t=x}(y)\mathrm{d}P_{X_t}(x).$$

Theorem. Let $(\mu_t(\cdot|y), v_t^y)$ satisfy the (CE), then the following (μ_t, v_t) also satisfy the (CE):

$$\mu_t = \pi^1_\#(\mu_t(\cdot|y) \times_y \mu_1), \qquad p_t(x) = \int p_t^y(x)\mathrm{d}\mu_1(y) \tag{1}$$

$$v_t(x) = \int_{\mathbb{R}^d} v_t^y(x)\,\mathrm{d}P_{X_1|X_t=x}(y) \tag{2}$$

Rewriting the loss function:

$$\mathcal{L}(\theta) := \mathbb{E}_{t\sim\mathcal{U}(0,T),x\sim\mu_t}\left[\left\|v_t^\theta(x) - v_t(x)\right\|^2\right] = \mathbb{E}\left[\|v_t^\theta\|^2\right] - 2\mathbb{E}\left[\langle v_t, v_t^\theta\rangle\right] + \underbrace{\mathbb{E}\left[\|v_t\|^2\right]}_{\text{const}}$$

Consider first and second term separately.

1.

$$\begin{aligned}\mathbb{E}_{(t,x)\sim\mu_t\times_t\mathrm{d}t}\left[\|v_t^\theta(x)\|^2\right] &= \int_0^1\int_{\mathbb{R}^d}\|v_t^\theta(x)\|^2\mathrm{d}\mu_t(x)\mathrm{d}t\\ &= \int_0^1\int_{\mathbb{R}^d}\int_{\mathbb{R}^d}\|v_t^\theta(x)\|^2\mathrm{d}P_{X_1|X_t=x}(y)\mathrm{d}P_{X_t}(x)\mathrm{d}t\\ &= \int_0^1\int_{\mathbb{R}^d\times\mathbb{R}^d}\|v_t^\theta(x)\|^2\mathrm{d}P_{X_t,X_1}(x,y)\mathrm{d}t\\ &= \int_0^1\int_{\mathbb{R}^d\times\mathbb{R}^d}\|v_t^\theta(x)\|^2\mathrm{d}P_{X_t|X_1=y}(x)\mathrm{d}P_{X_1}(y)\mathrm{d}t\\ &= \mathbb{E}_{t\sim\mathcal{U}(0,T),\,y\sim\mu_1,\,x\sim\mu_t(\cdot|y)}\left[\|v_t^\theta(x))\|^2\right]\end{aligned}$$

2.

$$\begin{aligned}\mathbb{E}_{(t,x)\sim\mu_t\times_t\mathrm{d}t}\left[\langle v_t^\theta(x),v_t(x)\rangle\right] &= \int_0^1\int_{\mathbb{R}^d}\langle v_t^\theta(x),\int_{\mathbb{R}^d}v_t^y(x)\mathrm{d}P_{X_1|X_t=x}(y)\rangle\mathrm{d}P_{X_t}(x)\mathrm{d}t\\ &= \int_0^1\int_{\mathbb{R}^d\times\mathbb{R}^d}\langle v_t^\theta(x),v_t^y(x)\rangle\mathrm{d}P_{X_t,X_1}(x,y)\mathrm{d}t\\ &= \mathbb{E}_{t\sim\mathcal{U}(0,T),\,y\sim\mu_1,\,x\sim\mu_t(\cdot|y)}\left[\langle v_t^\theta(x),v_t^y(x)\rangle\right]\end{aligned}$$

Velocity Fields from Couplings

How we can obtain conditioned velocity fields?

- from couplings/linear processes
- from special processes:

  Duong, Chemseddine, Friz, Steidl: Telegrapher's generative model via Kac flows

  Chemseddine, Kornhardt, Duong, Steidl: Adapting noise to data: generative flows from 1D processes

Conditioned velocity fields from couplings

Curve ends: $\mu_0 = \mu_Z,\ \mu_1 = \mu_X$

Arbitrary coupling: $\alpha \in \Pi(\mu_0, \mu_1),\ (X_0, X_1) \sim \alpha$

Interpolation map: $e_t(x, y) = (1-t)x + ty,\ t \in [0, 1]$

Theorem The pair (μ_t, v_t) induced by α, i.e.

$$\mu_t := e_{t,\sharp}\alpha \qquad X_t = (1-t)X_0 + tX_1 \sim \mu_t$$

$$v_t\mu_t := e_{t,\sharp}[(y-x)\alpha] \quad \text{for a.e. } t \in [0,1].$$

satisfies the (CE). Further,

$$\mu_t^y = e_{t,\sharp}(P_{X_t|X_1=y} \times \delta_y),$$

$$v_t^y = \frac{y-x}{1-t}$$

fulfill the (CE) and are related to μ_t and v_t by (1) and (2), resp.

Example

Example: Let $\alpha = \mu_0 \times \mu_1$ and $\mu_0 = \mathcal{N}(0, I_d)$. Then we show on the next slight that the pair (μ_t, v_t) with

$$\mu_t := e_{t,\sharp}\alpha, \qquad v_t\mu_t := e_{t,\sharp}[(y-x)\alpha]$$

is given by $\mu_t = p_t \mathrm{d}x$:

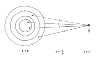
$$p_t(x) = C_t \int_{\mathbb{R}^d} \mathrm{e}^{-\frac{\|x-ty\|^2}{2(1-t)^2}} \mathrm{d}\mu_1(y) \quad \text{with} \quad C_t := \left(2\pi(1-t)^2\right)^{-\frac{d}{2}}$$

$$v_t = \frac{1-t}{t} \underbrace{\nabla \log p_t}_{\text{score}} + \frac{x}{t}$$

and thus for $\mu_1 = \delta_y$,

$$p_t(x|y) = C_t\, \mathrm{e}^{-\frac{\|x-ty\|^2}{2(1-t)^2}}$$

$$v_t^y(x) = \frac{y-x}{1-t}$$



and $\mu_t(\cdot|y) = T^y_{t,\sharp}\mu_0 \quad$ with $\quad T_t^y(x) := (1-t)x + ty$

ODE with $T_t^y = \phi_t$:

$$\partial_t T_t^y(x) = v_t^y(T_t^y(x)) = \frac{y - ((1-t)x + ty)}{1-t} = y - x$$

Example

1. For $\mu_t := e_{t,\sharp}\alpha$ we get with $x := (z - yt)/(1-t)$:

$$\int \varphi \,\mathrm{d}\mu_t = \int \varphi \,\mathrm{d}e_{t,\sharp}\alpha = \int \varphi\left(e_t(x,y)\right)\,\mathrm{d}\alpha(x,y)$$

$$= (2\pi)^{-\frac{d}{2}} \int \varphi\left((1-t)x + ty\right) \mathrm{e}^{-\frac{\|x\|^2}{2}} \,\mathrm{d}x\mathrm{d}\mu_1(y),$$

$$= \left(2\pi(1-t)^2\right)^{-\frac{d}{2}} \int \varphi(z)\mathrm{e}^{-\frac{\|z-ty\|^2}{2(1-t)^2}} \,\mathrm{d}\mu_1(y)\mathrm{d}z.$$

Example continued

2. Further we verify for $f_t := \frac{1-t}{t}\log p_t + \frac{1}{2t}\|\cdot\|^2$ and $\mu_t = p_t \mathrm{d}x$ that

$$\begin{aligned}
\int_{\mathbb{R}^d} \varphi\, \nabla f_t \,\mathrm{d}\mu_t &= \tfrac{1-t}{t}\int_{\mathbb{R}^d} \varphi(x)\frac{\nabla p_t(x)}{p_t(x)} p_t(x)\,\mathrm{d}x + \frac{1}{t}\int_{\mathbb{R}^d} \varphi(x)\, x\, p_t(x)\,\mathrm{d}x \\
&= \frac{1}{t(1-t)^{d+1}(2\pi)^{\frac{d}{2}}}\int_{\mathbb{R}^d\times\mathbb{R}^d} \varphi(x)(ty-x)\mathrm{e}^{-\frac{\|x-ty\|^2}{2(1-t)^2}}\,\mathrm{d}\mu_1(y)\,\mathrm{d}x \\
&\quad + \frac{1}{t(1-t)^{d}(2\pi)^{\frac{d}{2}}}\int_{\mathbb{R}^d\times\mathbb{R}^d} \varphi(x) x \mathrm{e}^{-\frac{\|x-ty\|^2}{2(1-t)^2}}\,\mathrm{d}\mu_1(y)\mathrm{d}x \\
&= \frac{1}{(1-t)^{d+1}(2\pi)^{\frac{d}{2}}}\int_{\mathbb{R}^d\times\mathbb{R}^d} \varphi(x)(y-x)\mathrm{e}^{-\frac{\|x-ty\|^2}{2(1-t)^2}}\,\mathrm{d}\mu_1(y)\,\mathrm{d}x.
\end{aligned}$$

On the other hand

$$\begin{aligned}
\int_{\mathbb{R}^d} \varphi\, v_t \,\mathrm{d}\mu_t &= \int_{\mathbb{R}^d} \varphi \,\mathrm{d}e_{t,\sharp}(y-x)\alpha \\
&= \frac{1}{(2\pi)^{\frac{d}{2}}}\int_{\mathbb{R}^d\times\mathbb{R}^d} \varphi(e_t(x,y))(y-x)\mathrm{e}^{-\frac{\|x\|^2}{2}}\,\mathrm{d}\mu_1(y)\mathrm{d}x \\
&= \frac{1}{(1-t)^{d+1}(2\pi)^{\frac{d}{2}}}\int_{\mathbb{R}^d\times\mathbb{R}^d} \varphi(z)(y-z)\mathrm{e}^{-\frac{\|z-ty\|^2}{2(1-t)^2}}\,\mathrm{d}\mu_1(y)\mathrm{d}z
\end{aligned}$$

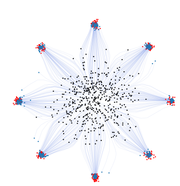
Loss function

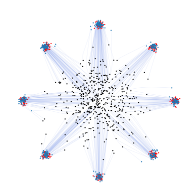
Recall $\mu_t = P_{X_t} = (e_t)_\# \mu_0 = (e_t)_\# P_{X_0}$, $\alpha = P_{X_0,X_1}$, $v_t^y = \frac{y-x}{1-t}$

$$
\begin{aligned}
\mathcal{L}(\theta) &:= \mathbb{E}_{t\sim\mathcal{U}(0,T),\, y\sim\mu_1,\, x\sim\mu_t(\cdot|y)} \left[\left\| v_t^\theta(x) - v_t^y(x) \right\|^2 \right] \\
&= \int_0^1 \int_{\mathbb{R}^d\times\mathbb{R}^d} \left\| v_t^\theta(x) - v_t^y(x) \right\|^2 \mathrm{d}P_{X_t|X_1=y}(x)\mathrm{d}P_{X_1}(y)\mathrm{d}t \\
&= \int_0^1 \int_{\mathbb{R}^d\times\mathbb{R}^d} \left\| v_t^\theta(x) - v_t^y(x) \right\|^2 \mathrm{d}P_{X_t,X_1}(x,y)\mathrm{d}t \\
&= \int_0^1 \int_{\mathbb{R}^d\times\mathbb{R}^d} \left\| v_t^\theta(e_t(x,y)) - v_t^y(e_t(x,y)) \right\|^2 \mathrm{d}P_{X_0,X_1}(x,y)\mathrm{d}t \\
&= \mathbb{E}_{t\sim\mathcal{U}(0,T),\, (x,y)\sim\alpha} \left[\| v_t^\theta(e_t(x,y)) - (y-x) \|^2 \right]
\end{aligned}
$$

All you need is a coupling α !

- independent coupling: $\alpha = \mu_0 \times \mu_1$
- optimal coupling: α minimizer of $W_2^2(\mu_0,\mu_1)$ with advantage that $\alpha = (\mathrm{Id}, T)_\sharp \mu_0$ is induced (often) by a Monge map $T : \mathbb{R}^d \to \mathbb{R}^d$ and we have straight trajectories $\phi_t(x) = T_t(x) := (1-t)x + tT(x)$





Let $X_t \in L^2(\Omega, \mathbb{R}^d, \mathbb{P})$ be continuously differentiable in every $t \in [0,1]$. Let

$$\mu_t := X_{t,\sharp}\mathbb{P} = P_{X_t}, \quad \text{and} \quad v_t := \mathbb{E}[\partial_t X_t | X_t = \cdot].$$

Theorem. Under some assumptions (μ_t, v_t) fulfills the continuity equation

Example: Let $X_t = (1-t)X_0 + tX_1$ for independent random variables X_0, X_1, where $X_0 \sim \mathcal{N}(0, I_d)$. Then we know already

$$P_{X_t|X_1=y} = C_t \, \mathrm{e}^{-\frac{\|x-ty\|^2}{2(1-t)^2}} \mathrm{d}x \sim \mu_t(\cdot|y)$$

Further we get with $\mathbb{E}[X_t|X_t] = X_t$ that

$$v_t(x) = \mathbb{E}[\partial_t X_t|X_t = x] = \mathbb{E}[X_1 - X_0|X_t = x] = \mathbb{E}\left[\frac{X_1}{1-t} - \frac{X_t}{1-t}\middle|X_t = x\right]$$
$$= \frac{1}{1-t}\mathbb{E}\left[X_1\middle|X_t = x\right] - \frac{x}{t-1}$$

Using Tweedie's (Miyasawa's) formula

$$\nabla_x \log p_t(x) = \frac{t\mathbb{E}[X_1|X_t = x] - x}{(1-t)^2}$$
$$\mathbb{E}[X_1|X_t = x] = \frac{(1-t)^2}{t}\nabla_x \log p_t(x) + \frac{x}{t}$$

we finally obtain

$$v_t(x) = \frac{1-t}{t}\nabla_x \log p_t(x) + \frac{x}{t(1-t)} - \frac{x}{1-t}$$
$$= \frac{1-t}{t}\nabla_x \log p_t(x) + \frac{x}{t},$$

Remark: Tweedie's Formula

Consider

$$y = x + \eta, \qquad \eta \sim \mathcal{N}(0, \sigma^2 I)$$

The observation density is related to the prior via Bayes' Theorem:

$$p(y) = \int p(y|x)p(x)\,\mathrm{d}x = \int g(y-x)p(x)\,\mathrm{d}x, \quad g(z) := \frac{1}{(2\pi\sigma^2)^{N/2}} e^{-\|z\|^2/(2\sigma^2)}.$$

For a given observation y, the minimum mean squared error (MMSE) estimator is given as

$$\hat{x}(y) = \int x\, p(x|y)\,\mathrm{d}x = \int x \frac{p(y|x)p(x)}{p(y)}\,\mathrm{d}x.$$

The MMSE can be computed using Tweedie's (Miyasawa's) formula

$$\hat{x}(y) = y + \sigma^2 \nabla_y \log p(y),$$

which can be seen directly by computing

$$\nabla_y p(y) = \frac{1}{\sigma^2} \int (x-y) g(y-x) p(x)\,\mathrm{d}x = \frac{1}{\sigma^2} \int (x-y) p(y,x)\,\mathrm{d}x,$$

and then multiplying both sides by $\sigma^2/p(y)$ gives

$$\frac{\sigma^2 \nabla_y p(y)}{p(y)} = \int x\, p(x|y)\,\mathrm{d}x - \int y\, p(x|y)\,\mathrm{d}x = \hat{x}(y) - y.$$

2. Generative Flows

Forward SDE computes $(X_t)_{t\in[0,T]}$ by

$$dX_t = f(t, X_t)\,dt + g(t)\,dW_t, \quad X_0 \sim P_{\text{data}},$$

where W_t is a standard Brownian motion.

Example: Usual choice is

$$f(t, X_t) := -\frac{1}{2}\beta_t X_t \quad \text{and} \quad g(t) := \sqrt{\beta_t}$$

with a positive, increasing so-called "time schedule" β_t. Then SDE becomes linear

$$dX_t = -\frac{1}{2}\beta_t X_t\,dt + \sqrt{\beta_t}\,dW_t, \quad X_0 \sim P_{\text{data}},$$

which has the closed form solution

$$X_t = \sqrt{1 - e^{-h(t)}}Z + e^{\frac{-h(t)}{2}}X_0, \quad h(t) := \int_0^t \beta_s\,ds, \qquad Z \sim \mathcal{N}(0, I_d)$$

Hence, X_t reaches Z for $t \to \infty$.

Reverse SDE becomes

$$dY_t = \Big(- f(T-t, Y_t) + g(T-t)^2 \underbrace{\nabla \log p_{X_{T-t}}(Y_t)}_{\text{score}}\Big)\,dt + g(T-t)\,dW_t, \quad Y_0 \sim X_T.$$

Approximation of score by a neural network s^θ:

$$\begin{aligned}\mathcal{L}(\theta) &:= \mathbb{E}_{t\sim\mathcal{L}_{[0,T]}}\mathbb{E}_{x\sim P_{X_t}}\left[\|s_t^\theta(x) - \nabla \log p_{X_t}(x)\|^2\right] \\ &= \mathbb{E}_{t\sim\mathcal{L}_{[0,T]},x_0\sim X_0,x\sim P_{X_t|X_0=x_0}}\left[\|s_t^\theta(x) - \nabla \log p_{X_t|X_0=x_0}(x)\|^2\right] + \text{const}\end{aligned}$$

Example: (continued)

$$P_{X_t|X_0=x_0} = \mathcal{N}\left(b_t\, x_0,\, (1-b_t^2)\, I_d\right) \quad \text{with} \quad b_t = \mathrm{e}^{-\frac{h(t)}{2}}$$

This implies

$$\nabla \log p_{X_t|X_0=x_0}(x) = \nabla\left(-\tfrac{1}{2(1-b_t^2)}\|x - b_t x_0\|^2\right) = -\frac{x - b_t x_0}{1-b_t^2}.$$

Plugging this into the loss function, we get

$$\mathcal{L}(\theta) = \mathbb{E}_{t\sim\mathcal{L}_{[0,T]},x_0\sim X_0,x\sim P_{X_t|X_0=x_0}}\left[\left\|s_t^\theta(x) + \frac{x - b_t x_0}{1-b_t^2}\right\|^2\right].$$

Once the score is computed, we can use it in the reverse SDE starting with the $Z \sim \mathcal{N}(0, I_d)$ which approximates X_T for T large enough.

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27

For the forward PDE

$$\mathrm{d}X_t = f(t, X_t)\,\mathrm{d}t + g(t)\,\mathrm{d}W_t, \quad X_0 \sim P_{\mathrm{data}},$$

the corresponding densities $p_t = p_{X_t}$ fulfill the Fokker-Planck equation (CE)

$$\begin{aligned}\partial_t p_t(x) &= -\nabla \cdot (f(t,x)p_t(x)) + \frac{1}{2}g(t)^2 \Delta p_t(x)\\ &= -\nabla \cdot \Big(\underbrace{\big(f(t,x) - \frac{1}{2}g(t)^2 \nabla_x \log p_t(x)\big)}_{v_t} p_t(x) \Big)\end{aligned}$$

Wasserstein gradient flow for minimizing $\mathcal{F} : \mathcal{P}_2(\mathbb{R}^d) \to \mathbb{R}$

Choosing $v_t \in -\partial \mathcal{F}(\mu_t) = -\nabla \delta \mathcal{F}(\mu_t)$ leads to the (CE)

$$\partial_t p_t = \nabla \cdot (p_t \, \nabla \delta \mathcal{F}(\mu_t))$$

- $\mathcal{F}_\nu(\mu) = \mathrm{KL}(\mu, \nu) = \int_{\mathbb{R}^d} \log p(x) p(x)\,\mathrm{d}x - \int_{\mathbb{R}^d} \log q(x) p(x)\,\mathrm{d}x$, $\mu = p\lambda$, $\nu = q\lambda$
  Then
  $$\delta \mathcal{F}(p_t) = 1 + \log p_t - \log q, \quad \nabla \delta \mathcal{F}(p_t) = \frac{1}{p_t}\nabla p_t - \log q$$
  so that
  $$\partial_t p_t = \Delta p_t - \nabla \cdot (p_t \nabla \log q)$$
  which is the above Fokker-Planck equation for $f(t,x) = \nabla \log q(x)$ and $g(t) = \sqrt{2}$.

Fokker Planck Equation

For $q(x) = C^{-1}\mathrm{e}^{-\beta\Psi(x)}$ we obtain the

Overdamped Langevin SDE:

$$X(0) = X^0 \sim p^0$$
$$\mathrm{d}X_t = -\nabla\Psi(X_t)\mathrm{d}t + \sqrt{2\beta^{-1}}\mathrm{d}W_t$$

Euler-Maruyama forward step with step size η:

$$X_k := X_{k-1} - \eta\nabla\Psi(X_{k-1}) + \sqrt{2\beta^{-1}\eta}\,\xi_k, \quad \xi_k \sim \mathcal{N}(0, I_d)$$

Hint: Use substitution $s = \frac{t}{\beta}$, $\sqrt{\beta^{-1}}W_t = W_{t/\beta}$

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MATH+

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Oberwolfach Seminars
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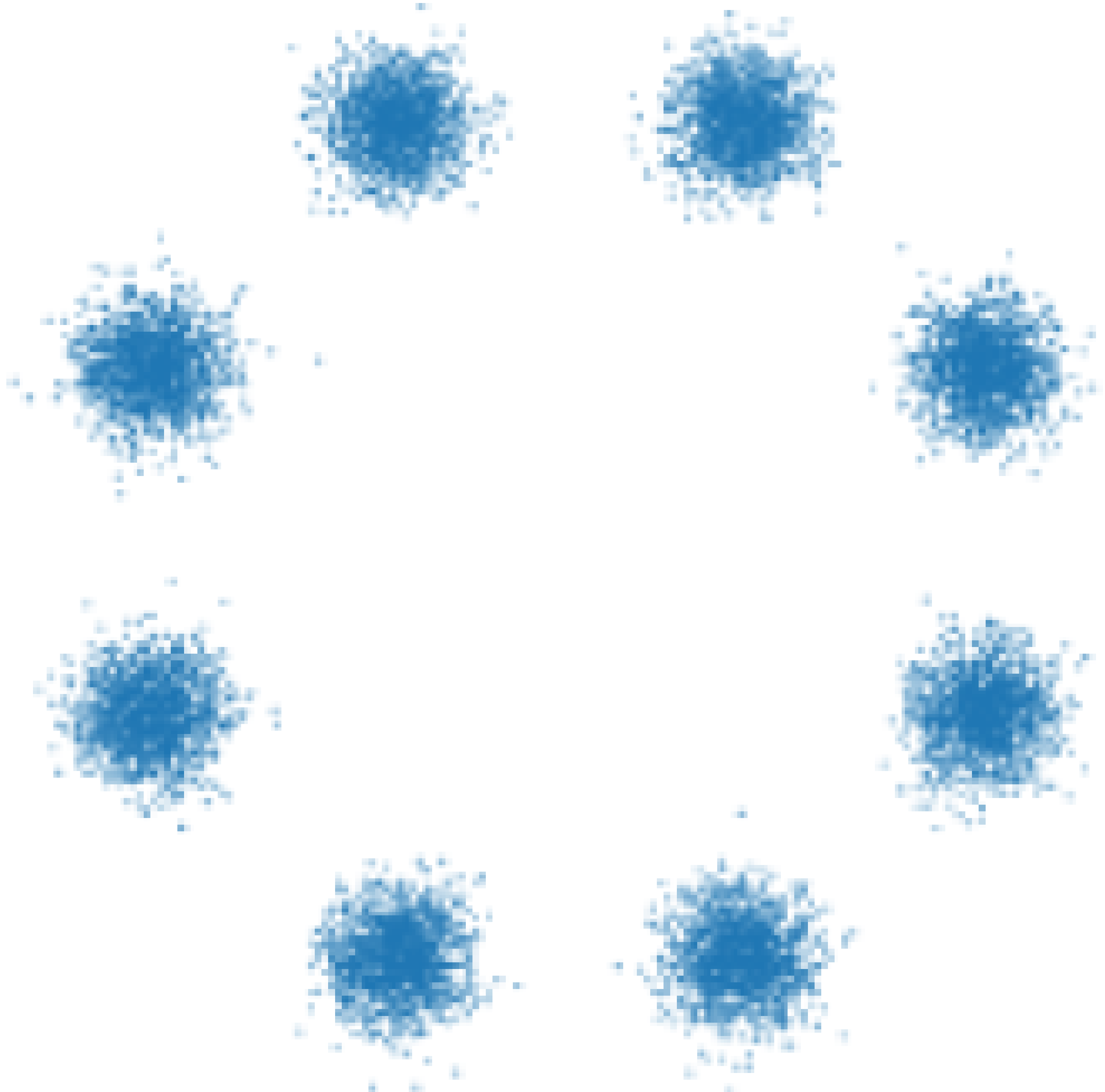
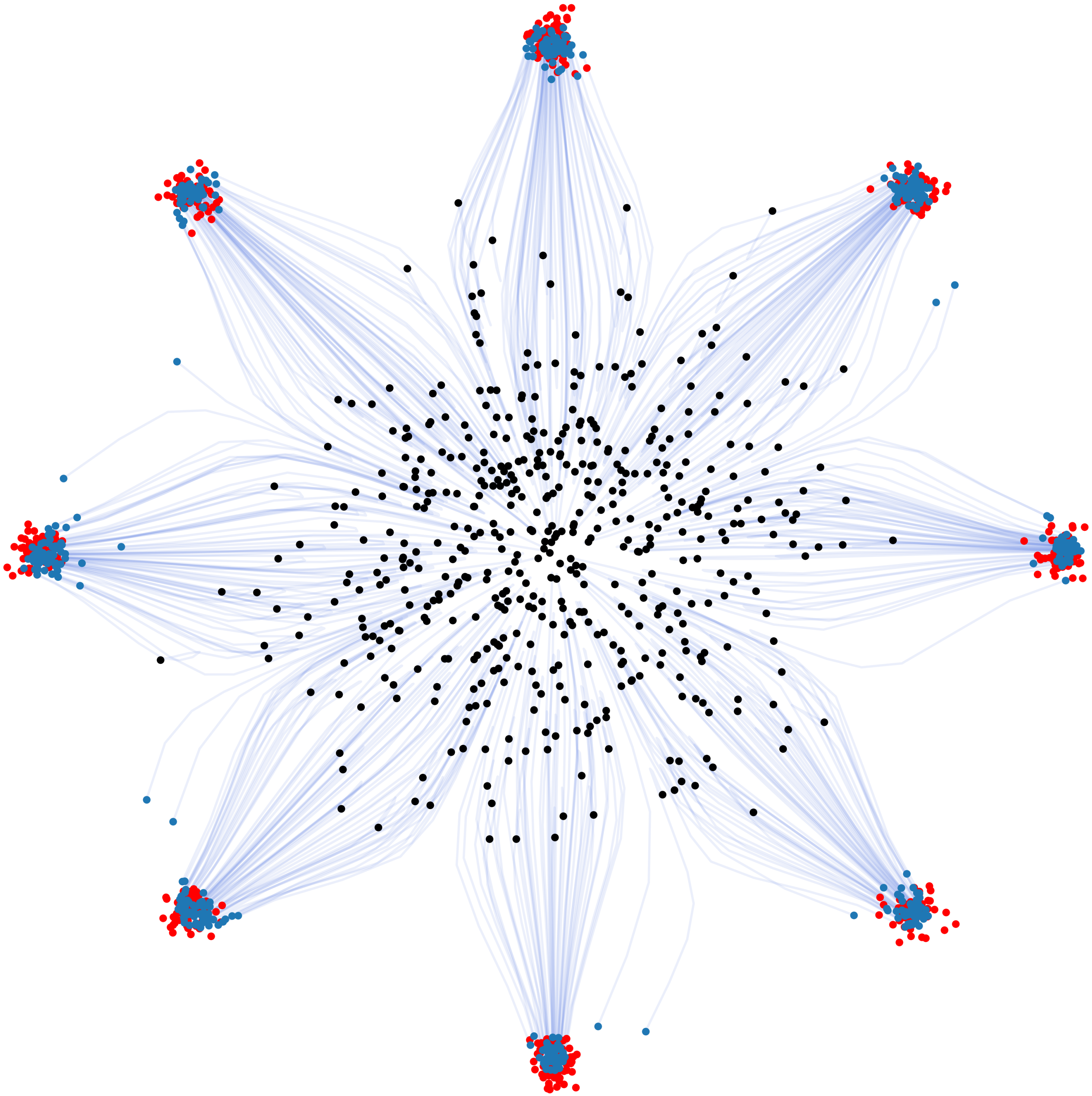
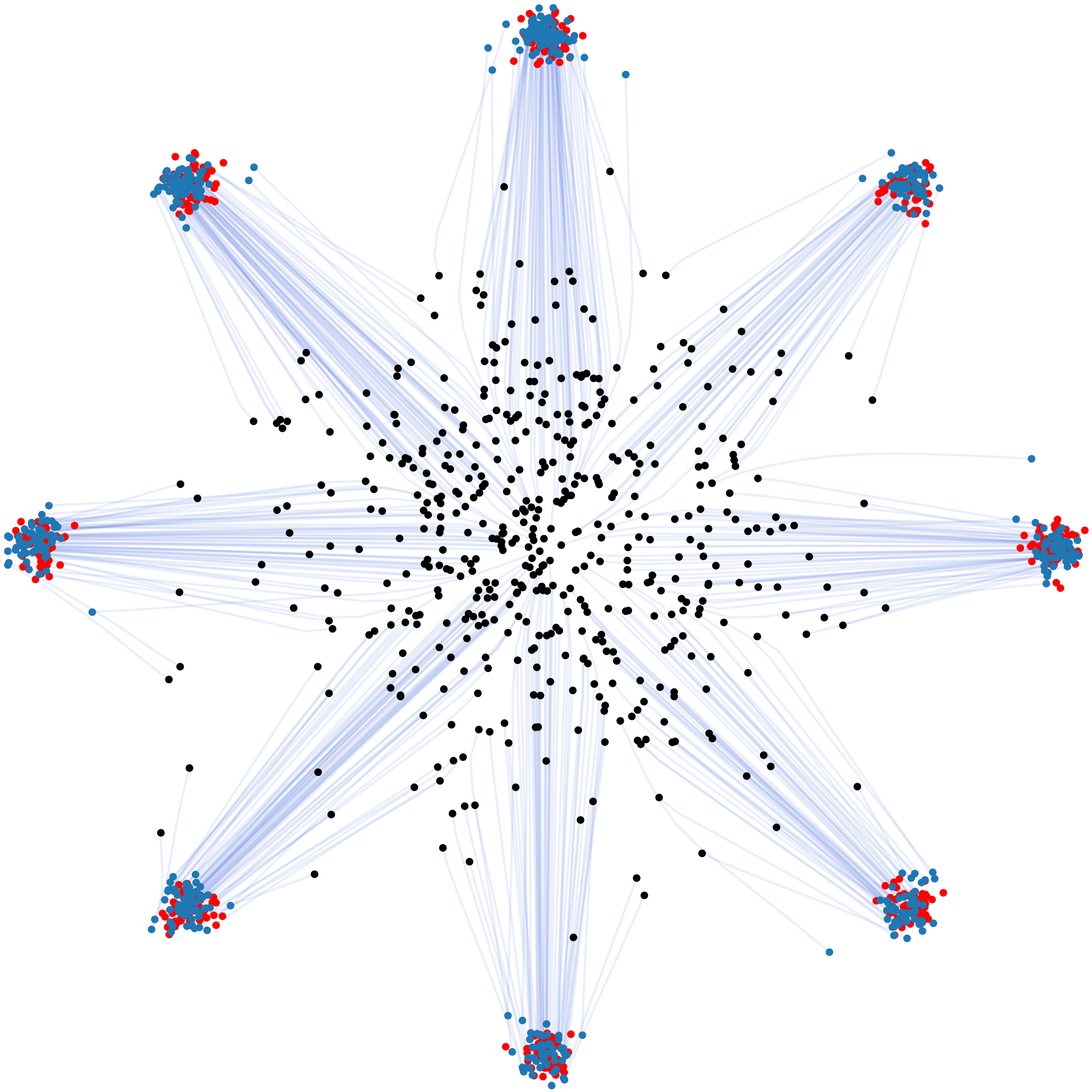
Variational and Information Flows in Machine Learning and Optimal Transport

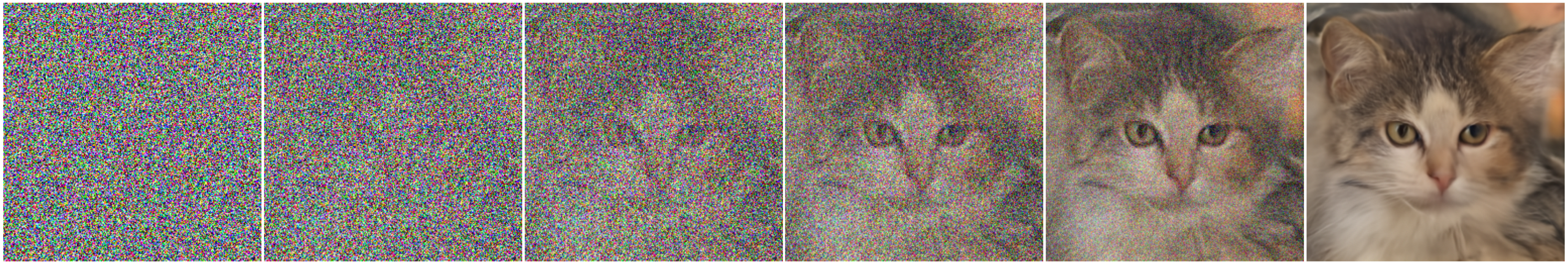
Wuchen Li
Bernhard Schmitzer
Gabriele Steidl
François-Xavier Vialard
Christian Wald

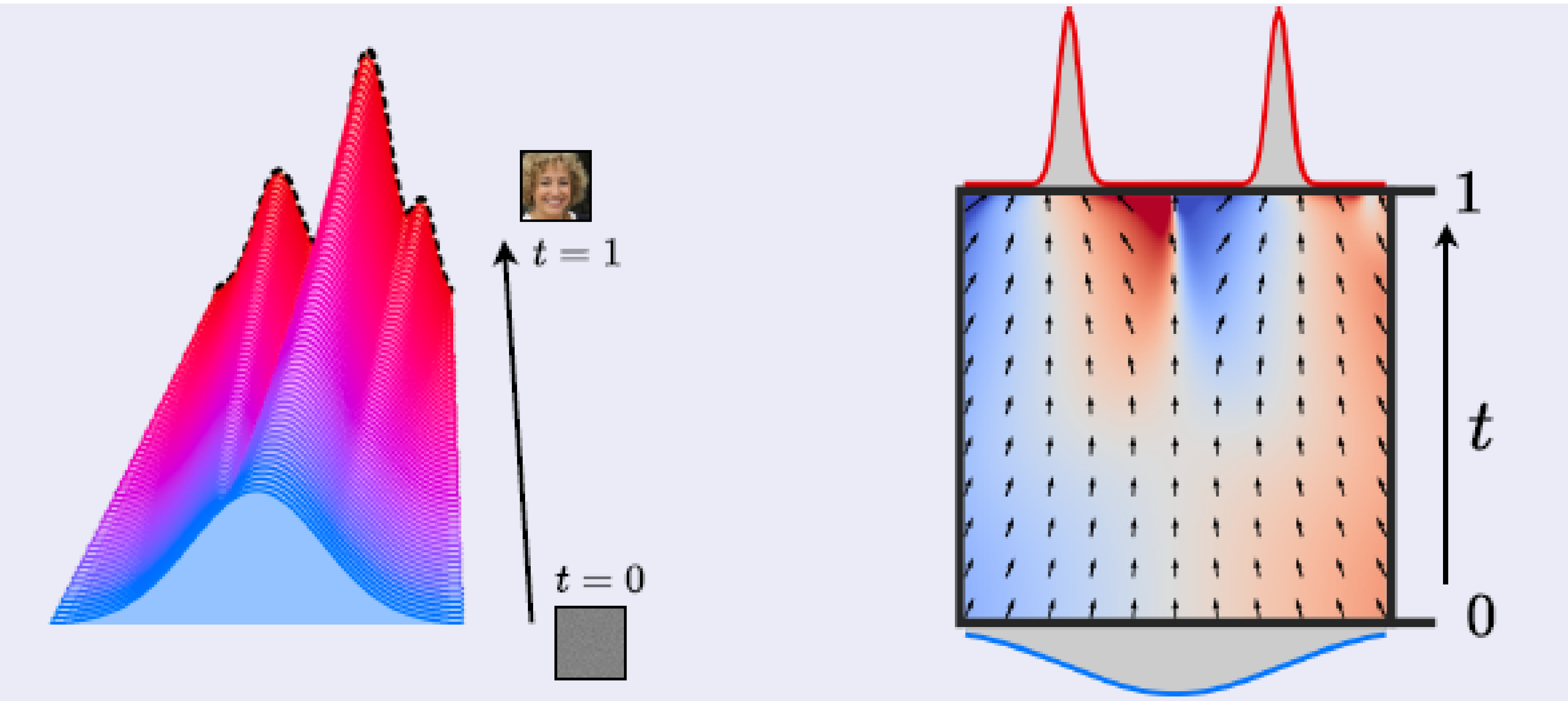
Birkhäuser











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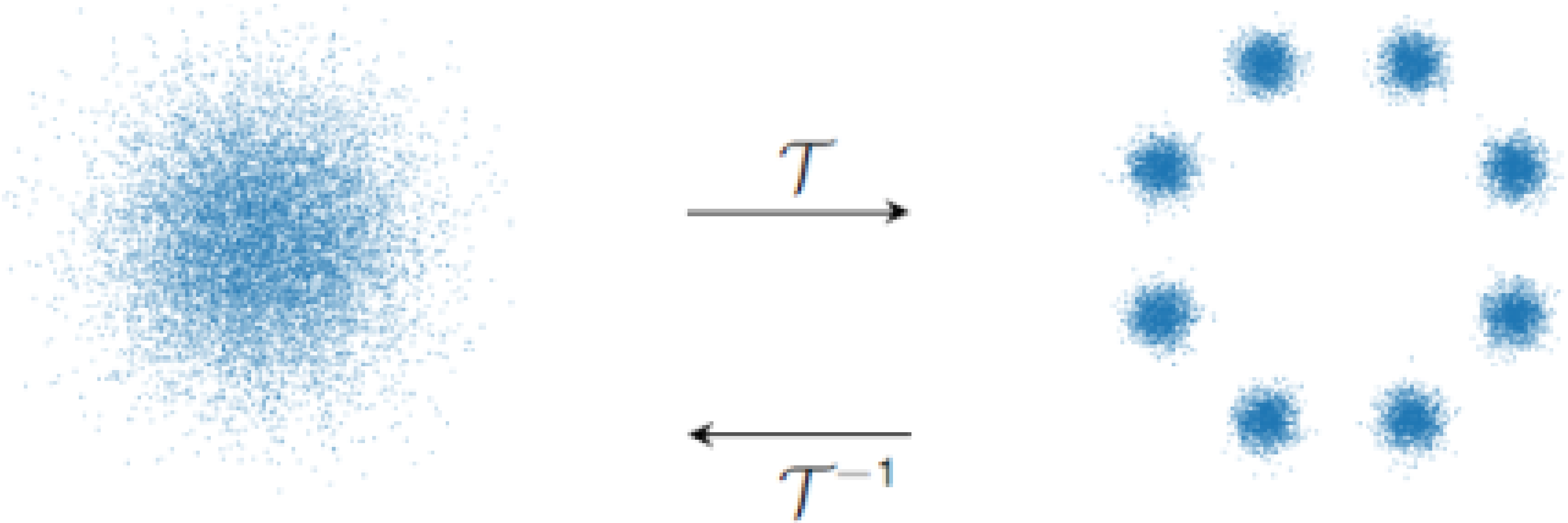
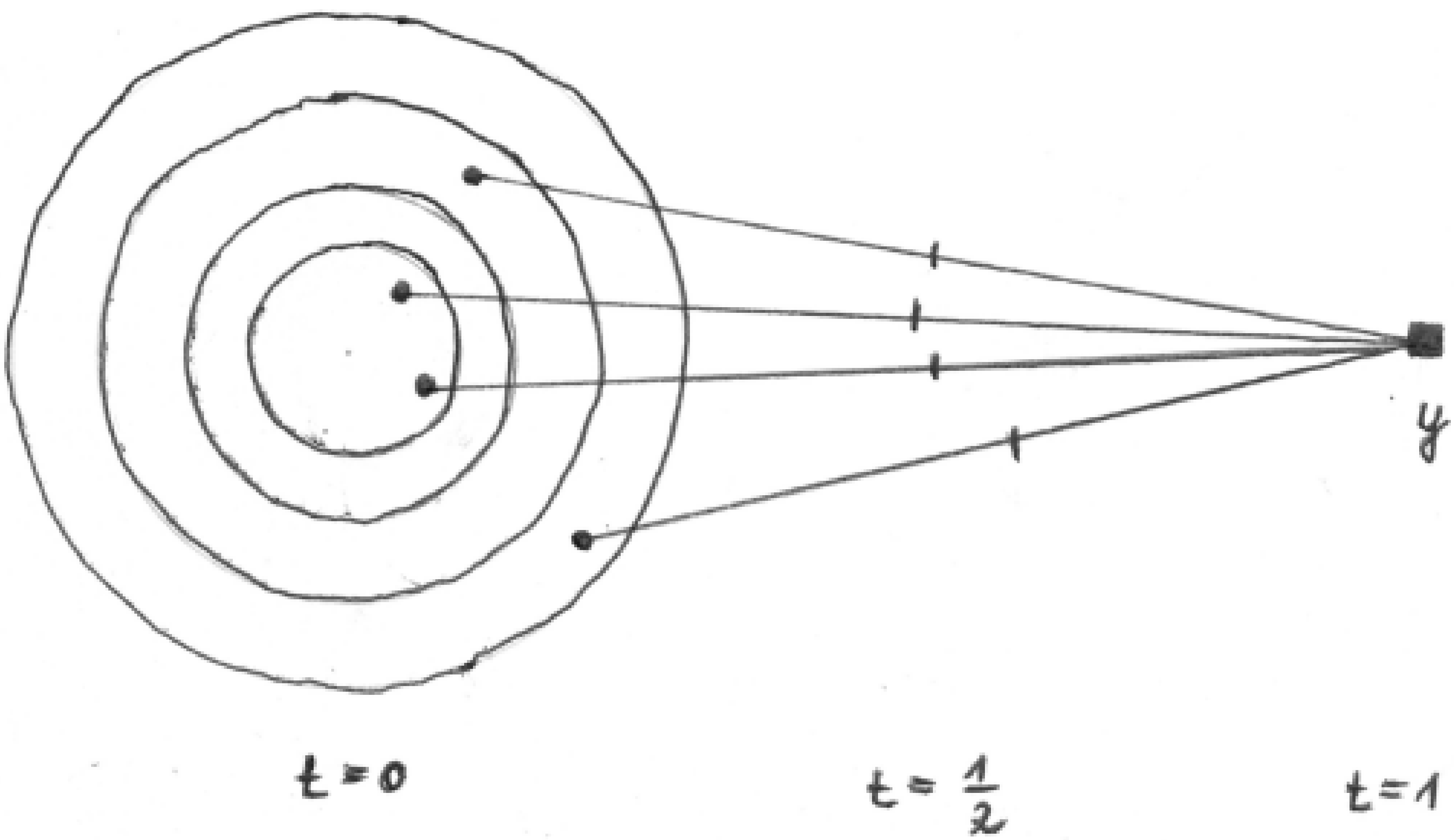
$t = 0$
1
t
0


$\mathcal{T}$
$\mathcal{T}^{-1}$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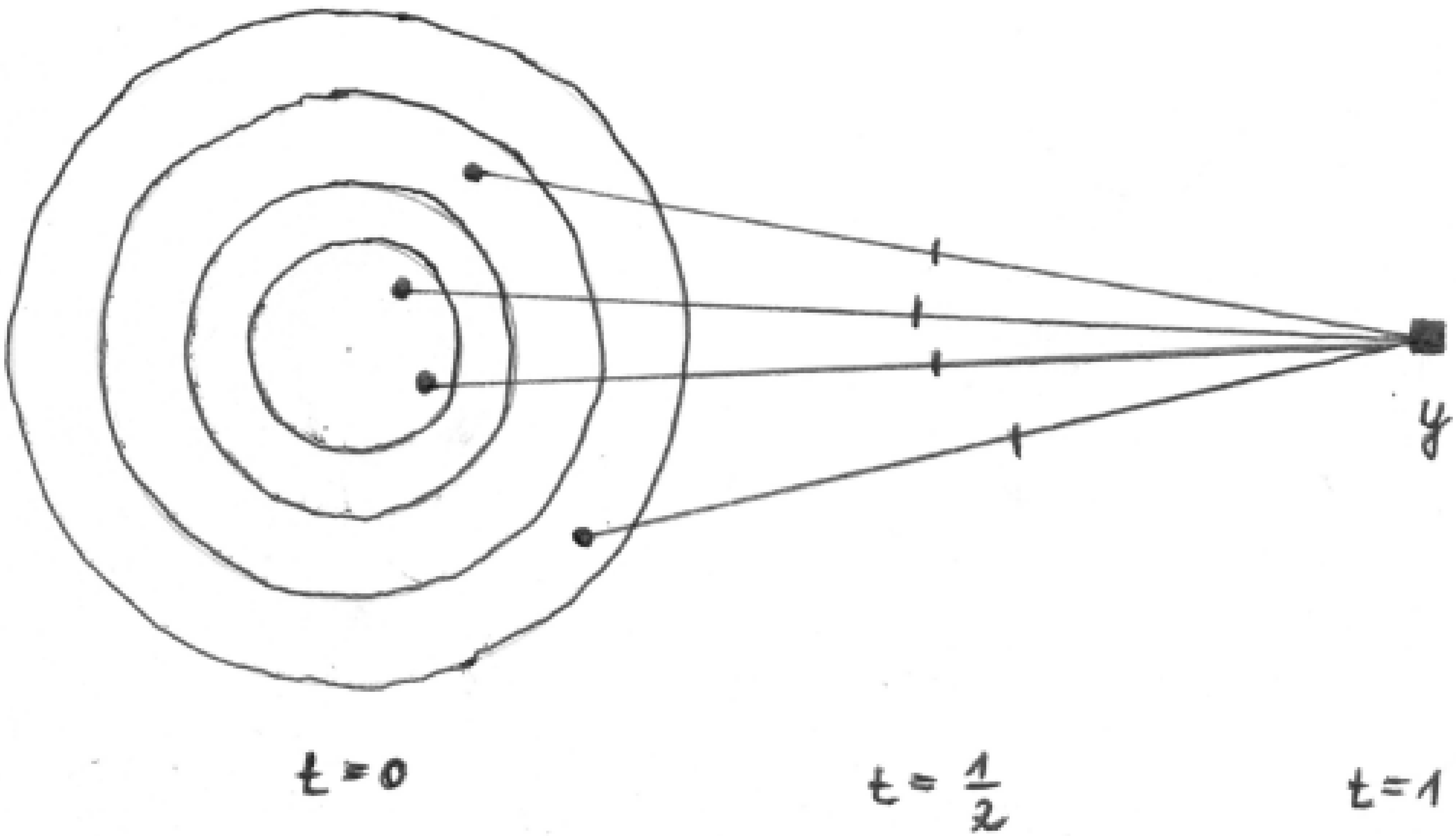
t = 1/2
t = 1


y
t = 0
t = 1/2
t = 1